2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.241 Introduction to Computer Graphics

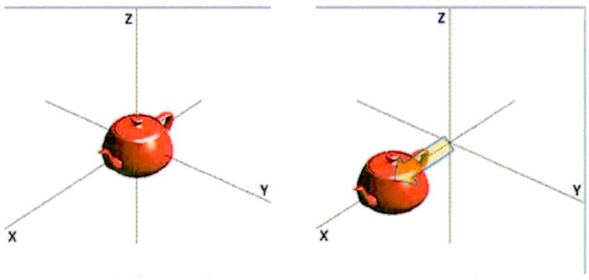
Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

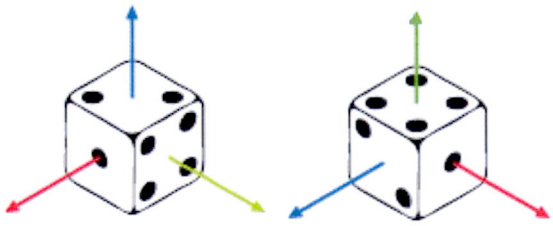
Question 1

1.a Write the sequence of OpenGL 3D transformations that maps the given shape on the left (see images below) to the shape on the right, for each of the two cases below. You can use a combination of the following functions: glTranslatef(tx, ty, tz), glRotatef(angle, x, y, z), and/or glScalef(sx, sy, sz).

i.



ii.



[3 marks]

1.b Give one example of an everyday 3D shape that would be represented well by each of the following 3D object representations. Explain your answer in each case.

i. Point cloud
ii. Parametric surface
iii. Voxels
iv. Scene graph

[4 marks]

1.c Write a vertex table and a face table for a cube (3D shape). The exact values of the vertices are not important as long as they form the overall shape.

[6 marks]

1.d You are given a ray with the equation P = (0,3,1) + t * (0,2,3), and a sphere with the equation $|P - (3,1,3)|^2 = 4^2$. Perform a ray-sphere intersection. Does the ray intersect with the sphere? If so, what is the closest value of t? Show your workings.

[7 marks]

Total 20 marks

Question 2

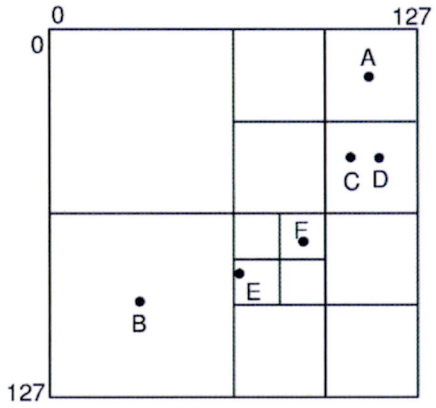
2.a Explain how the bounding volume hierarchy works for the purpose of rendering. Draw a diagram to show an example of the bounding volume hierarchy.

[4 marks]

2.b Give an example of gimbal lock and explain what it is. Draw a diagram to help with your explanation.

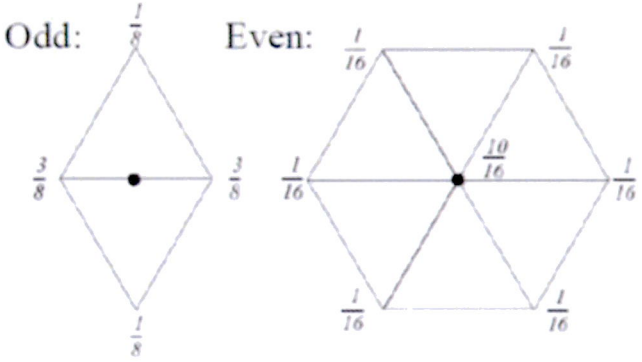
[4 marks]

2.c Draw a quadtree to represent the spatial arrangement given below. Indicate where the points A to F would be in your quadtree.



[6 marks]

2.d The rules below are for applying Loop Subdivision. We start with a triangular pyramid (made up of four triangles) and apply Loop Subdivision to the whole shape. After one iteration, how many faces are there in the result? In the process of applying this first iteration, how many times do we use each of the two rules below?



[6 marks]

Total 20 marks

Question 3

3.a Describe the concept of inverse kinematics by drawing a linked structure with 3 parts (and 3 joints). Clearly label each joint and any part that is important to inverse kinematics.

[4 marks]

3.b Explain the concept of convolution as applied to an image and give an example of its use. This concept of convolution for images is analogous to an important operation that was introduced for one of the 3D representations. Which representation is it and what is the operation?

[6 marks]

3.c Points A=(8,5,4) and B=(5,5,7) are points that form a parametric line. Is point C=(6,5,6) on this parametric line? If not, explain why not? If it is, what is the value of parameter t that gives point C in the parametric line AB?

[4 marks]

3.d You are given a cube made up of 12 triangles (i.e. two triangles per side of the cube). Describe how you would compute whether the cube intersect with an implicit surface.

[6 marks]

Total 20 marks

Question 4

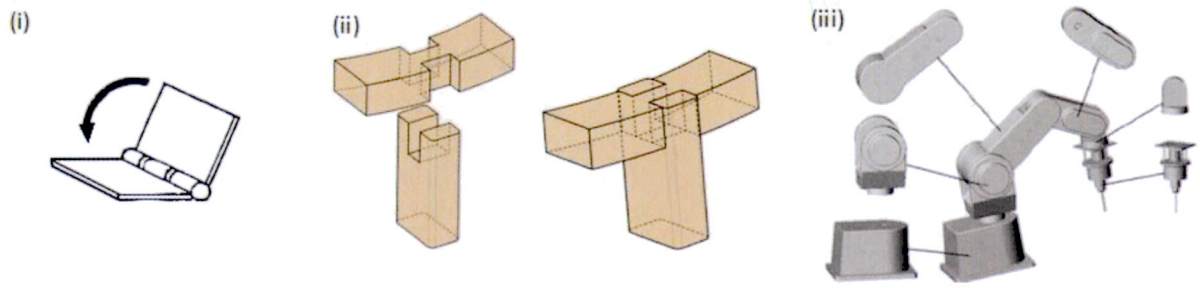
4.a Describe how mechanical motion capture works, and give one advantage and one disadvantage of this technique.

[4 marks]

4.b How would you generate a "flee" behaviour for a boids-like character? How would you generate an "arrival" behaviour?

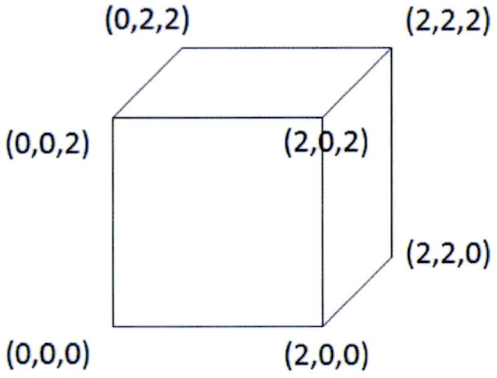
[4 marks]

4.c How many degrees of freedom does each of the following three cases have (assume that the bottom-most piece in each case stays fixed in position)? Explain your answers.



[6 marks]

4.d Apply Linear Subdivision to the cube below for the vertex (2,0,2). What is the updated position of this vertex after one iteration? Show all steps in your calculations.



[6 marks]

Total 20 marks

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